

Full Ansible Playbook — Ubuntu Web Stack Automation

This playbook automates:

- Apache
 - PHP
 - MySQL
 - Redis
 - Python
 - Flask support
 - Apache virtual hosts
 - Reverse proxy setup
 - Firewall (UFW)
 - SSH hardening
 - Apache hardening
 - Redis hardening
 - MySQL hardening
-

1. Inventory File

Create:

```
inventory.ini
```

Content:

```
[webservers]  
192.168.1.100 ansible_user=ubuntu
```

Replace:

- 192.168.1.100 with your VM IP
 - ubuntu with your SSH username
-

2. Install Required Collection

Run on control machine:

```
ansible-galaxy collection install community.mysql
```

3. Main Playbook

Create:

```
site.yml
```

Content:

```
---
- name: Ubuntu Web Stack Setup
  hosts: webserver
  become: yes

  vars:
    domain_name: example.local
    api_domain: api.local
    web_root: /var/www/example
    api_root: /var/www/api
    mysql_db: appdb
    mysql_user: appuser
    mysql_password: StrongPassword123!
    redis_password: StrongRedisPassword123!

  tasks:

    #####
    # UPDATE SYSTEM
    #####

    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Upgrade packages
      apt:
        upgrade: dist

    #####
    # INSTALL PACKAGES
    #####
```

```
- name: Install required packages
```

```
  apt:
```

```
    name:
```

- apache2
- mysql-server
- redis-server
- php
- libapache2-mod-php
- php-mysql
- php-redis
- php-cli
- php-curl
- php-xml
- php-mbstring
- python3
- python3-pip
- python3-venv
- python-is-python3
- ufw
- curl
- wget
- unzip
- git
- vim
- htop
- net-tools
- certbot
- python3-certbot-apache
- python3-pymysql

```
  state: present
```

```
#####
```

```
# APACHE MODULES
```

```
#####
```

```
- name: Enable Apache modules
```

```
  apache2_module:
```

```
    name: "{{ item }}"
```

```
    state: present
```

```
  loop:
```

- rewrite
- headers
- ssl
- proxy
- proxy_http

```
#####
```

```

# CREATE WEB DIRECTORIES
#####

- name: Create website directory
  file:
    path: "{{ web_root }}"
    state: directory
    owner: www-data
    group: www-data
    mode: '0755'

- name: Create API directory
  file:
    path: "{{ api_root }}"
    state: directory
    owner: www-data
    group: www-data
    mode: '0755'

#####
# CREATE TEST PAGES
#####

- name: Create index.html
  copy:
    dest: "{{ web_root }}/index.html"
    content: |
      <html>
      <body>
      <h1>Example Site</h1>
      </body>
      </html>

- name: Create PHP info page
  copy:
    dest: /var/www/html/info.php
    content: |
      <?php
      phpinfo();
      ?>

#####
# APACHE VIRTUAL HOST – WEBSITE
#####

- name: Configure website virtual host
  copy:
    dest: /etc/apache2/sites-available/example.conf

```

```

content: |
    <VirtualHost *:80>
        ServerName {{ domain_name }}
        DocumentRoot {{ web_root }}

        <Directory {{ web_root }}>
            AllowOverride All
            Require all granted
            Options -Indexes
        </Directory>

        ErrorLog ${APACHE_LOG_DIR}/example_error.log
        CustomLog ${APACHE_LOG_DIR}/example_access.log combined
    </VirtualHost>

#####
# APACHE VIRTUAL HOST – API REVERSE PROXY
#####

- name: Configure API virtual host
  copy:
    dest: /etc/apache2/sites-available/api.conf
    content: |
        <VirtualHost *:80>
            ServerName {{ api_domain }}

            ProxyPass / http://127.0.0.1:8000/
            ProxyPassReverse / http://127.0.0.1:8000/

            ErrorLog ${APACHE_LOG_DIR}/api_error.log
            CustomLog ${APACHE_LOG_DIR}/api_access.log combined
        </VirtualHost>

#####
# ENABLE SITES
#####

- name: Enable example site
  command: a2ensite example.conf
  args:
    creates: /etc/apache2/sites-enabled/example.conf

- name: Enable api site
  command: a2ensite api.conf
  args:
    creates: /etc/apache2/sites-enabled/api.conf

#####

```

```

# APACHE SECURITY HARDENING
#####

- name: Set ServerTokens
  lineinfile:
    path: /etc/apache2/conf-enabled/security.conf
    regexp: '^ServerTokens'
    line: 'ServerTokens Prod'

- name: Set ServerSignature
  lineinfile:
    path: /etc/apache2/conf-enabled/security.conf
    regexp: '^ServerSignature'
    line: 'ServerSignature Off'

- name: Disable TRACE method
  lineinfile:
    path: /etc/apache2/conf-enabled/security.conf
    regexp: '^TraceEnable'
    line: 'TraceEnable Off'

#####
# MYSQL HARDENING
#####

- name: Restrict MySQL to localhost
  lineinfile:
    path: /etc/mysql/mysql.conf.d/mysqld.cnf
    regexp: '^bind-address'
    line: 'bind-address = 127.0.0.1'

#####
# MYSQL DATABASE
#####

- name: Create MySQL database
  community.mysql.mysql_db:
    name: "{{ mysql_db }}"
    state: present
    login_unix_socket: /run/mysqld/mysqld.sock

- name: Create MySQL user
  community.mysql.mysql_user:
    name: "{{ mysql_user }}"
    password: "{{ mysql_password }}"
    priv: "{{ mysql_db }}.*:ALL"
    host: localhost
    state: present

```

```

login_unix_socket: /run/mysqld/mysqld.sock

#####
# REDIS HARDENING
#####

- name: Restrict Redis bind
  lineinfile:
    path: /etc/redis/redis.conf
    regexp: '^bind'
    line: 'bind 127.0.0.1'

- name: Enable Redis protected mode
  lineinfile:
    path: /etc/redis/redis.conf
    regexp: '^protected-mode'
    line: 'protected-mode yes'

- name: Set Redis password
  lineinfile:
    path: /etc/redis/redis.conf
    regexp: '^# requirepass'
    line: 'requirepass {{ redis_password }}'

#####
# FIREWALL CONFIGURATION
#####

- name: Allow OpenSSH
  ufw:
    rule: allow
    name: OpenSSH

- name: Allow HTTP
  ufw:
    rule: allow
    port: '80'
    proto: tcp

- name: Allow HTTPS
  ufw:
    rule: allow
    port: '443'
    proto: tcp

- name: Enable firewall
  ufw:
    state: enabled

```

```
#####  
# SSH HARDENING  
#####
```

- name: Disable root SSH login
lineinfile:
 - path: /etc/ssh/sshd_config
 - regexp: '^PermitRootLogin'
 - line: 'PermitRootLogin no'
- name: Disable password authentication
lineinfile:
 - path: /etc/ssh/sshd_config
 - regexp: '^PasswordAuthentication'
 - line: 'PasswordAuthentication no'
- name: Enable public key auth
lineinfile:
 - path: /etc/ssh/sshd_config
 - regexp: '^PubkeyAuthentication'
 - line: 'PubkeyAuthentication yes'

```
#####  
# PYTHON FLASK SETUP  
#####
```

- name: Install Flask
pip:
 - name: flask

```
#####  
# START SERVICES  
#####
```

- name: Start and enable Apache
systemd:
 - name: apache2
 - state: restarted
 - enabled: yes
- name: Start and enable MySQL
systemd:
 - name: mysql
 - state: restarted
 - enabled: yes
- name: Start and enable Redis


```

systemd:
  name: redis-server
  state: restarted
  enabled: yes

- name: Restart SSH
  systemd:
    name: ssh
    state: restarted

#####
# HOSTS FILE ENTRIES
#####

- name: Add example.local to hosts
  lineinfile:
    path: /etc/hosts
    line: '127.0.0.1 example.local'

- name: Add api.local to hosts
  lineinfile:
    path: /etc/hosts
    line: '127.0.0.1 api.local'

```

4. Run the Playbook

Test connectivity:

```
ansible webservers -i inventory.ini -m ping
```

Run playbook:

```
ansible-playbook -i inventory.ini site.yml
```

5. What This Playbook Configures

Apache

- Installs Apache

- Enables required modules
- Creates virtual hosts
- Enables reverse proxy
- Hardens security headers

PHP

- Installs PHP + modules
- Enables Redis/MySQL integration

MySQL

- Installs MySQL
- Creates database
- Creates user
- Restricts DB to localhost

Redis

- Installs Redis
- Restricts to localhost
- Enables protected mode
- Sets authentication password

Python

- Installs Python
- Installs Flask
- Enables python command alias

Security

- Configures firewall
 - Disables root SSH login
 - Disables SSH password authentication
 - Enables SSH key auth
 - Disables Apache version leakage
 - Disables directory listing
-

6. Notes

Certbot

Installed but not configured because:

- example.local
- api.local

are not publicly valid domains.

Use:

- real domain names OR
- self-signed certificates

for HTTPS testing.

7. Recommended Next Improvements

After learning this:

- Convert into Ansible roles
- Use Jinja2 templates
- Use Ansible Vault for passwords
- Add Gunicorn service automation
- Add Docker deployment
- Add CI/CD pipeline
- Add monitoring/logging
- Add fail2ban
- Add automatic backups